

Project 1: Grow a fern

Honor Code remainder:

Students are allowed to develop answers to specific problems together and check answers with each other. However, students are expected to write, document, understand, and submit their own work. At no point will copying and pasting the code of others be accepted or tolerated, especially code that has been submitted in previous semesters.

In this project, we'll discover how to create a stunning fern plot by employing an iterative process along with matrix multiplication.

Iteration and Matrix-Vector Multiplication

To iterate means to repeatedly apply the same rule or process. While this concept is straightforward and easy to code, it can lead to the creation of incredibly fascinating patterns, such as eyes, palm trees, and even galaxies.

Let take a look at simple examples of iteration. First, consider the rabbit rule "multiply by 6". If we start with a population of 10 and iterate we achieve 10, 60, 360, 2160, 12960, ... Second, if we instead apply the cannibal rule "multiply by 1/2" we arrive at 10, 5, 2.5, 1.25, 0.625, ...

To get more interesting patterns we move into higher dimensions. For example, instead of transforming one number into another via scalar multiplication we wish to transform a pair of numbers, say $x(1)$ and $x(2)$, into a second pair, say $y(1)$ and $y(2)$, via matrix multiplication. More precisely, we write

$$y(1) = A(1, 1) \cdot x(1) + A(1, 2) \cdot x(2) \quad (1)$$

$$y(2) = A(2, 1) \cdot x(1) + A(2, 2) \cdot x(2) \quad (2)$$

or more compactly

$$y = A * x,$$

where x and y are 2×1 vectors and A is 2×2 matrix of the form

$$A = \begin{bmatrix} A(1, 1) & A(1, 2) \\ A(2, 1) & A(2, 2) \end{bmatrix}.$$

For example, the matrix

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

rotates every vector clockwise by 90 degrees. Do you see it? Try it by hand for a few vectors. Then coax Matlab to do it by trying variations on the following sequence of commands:

```
A = [0 1; -1 0];
x = [1; 1];
y = A*x;
plot([0 x(1)], [0 x(2)]);
hold on
plot([0 y(1)], [0 y(2)], 'r');
y = A*y;
plot([0 y(1)], [0 y(2)], 'r--');
y = A*y;
plot([0 y(1)], [0 y(2)], 'r-.');
axis equal
```

Notice that since we want to iterate the process over and over, you can simply do: $y = A * y$. Moreover, since we are updating y each time, we need to use "hold on" to keep all vectors on the figure.

This example will draw a vector $\overrightarrow{(1,1)}$ by connecting the origin (0,0) and the point (1,1). However, for the project you just need to draw a single point, say $y = (1,1)$ as a dot. To plot a point y , you can use `plot(y(1),y(2),'.','MarkerSize',5)`.



Figure 1: Simple fern

1 Plotting a simple fern

This is the first part of your project. In this part, you will produce the exact plot as in Fig. (1).

Your task is to compute and plot the updated point z at each iteration.

As a strating point, use $z = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

In order, to update z at each iteration, you will generate each time a random number r that is uniformly distributed between 0 and 1.

1. In the case $r < 0.3994$, update z as following:

$$z = \begin{bmatrix} 0.4 & -0.3733 \\ 0.06 & 0.60 \end{bmatrix} z + \begin{bmatrix} 0.3533 \\ 0 \end{bmatrix}.$$

This is also means that we update z by the formula above with probability 0.3994.

2. In the case $r \geq 0.3994$, update z as following:

$$z = \begin{bmatrix} -0.8 & -0.1867 \\ 0.1371 & 0.80 \end{bmatrix} z + \begin{bmatrix} 1.1 \\ 0.1 \end{bmatrix}.$$



Figure 2: Advanced fern

This also means that we update z by the formula above with probability 0.6006.

Your next step is to experiment with the cutoff value 0.3994. What happens to the fern when you increase or decrease 0.3994? Try different values, and simply check your resulting plots.

To receive a full credit, you need to answer that question inside your script by using comments.

As an output for this part, produce a plot, where you use 4000 iterations and a fairly small markersize. Plot all points obtained through 4000 iterations. Include a title to your plot.

2 Plotting an advanced fern

This is the second part of your project.

We will be growing ferns by random repetition of a few simple matrix/vector transformations. Your task is to modify the existing code from part 1 and then to write and run code that produces a significantly more interesting fern as shown in Fig. (2). In particular, your new code should

begin with the same starting point as in part 1, then successively generate a random number and

1. Update

$$z = \begin{bmatrix} 0 & 0 \\ 0 & 0.16 \end{bmatrix} z$$

with probability 0.01.

2. Update

$$z = \begin{bmatrix} 0.85 & 0.04 \\ -0.04 & 0.85 \end{bmatrix} z + \begin{bmatrix} 0 \\ 1.6 \end{bmatrix}$$

with probability 0.75.

3. Update

$$z = \begin{bmatrix} 0.2 & -0.26 \\ 0.23 & 0.22 \end{bmatrix} z + \begin{bmatrix} 0 \\ 1.6 \end{bmatrix}$$

with probability 0.12.

4. Update

$$z = \begin{bmatrix} -0.15 & 0.28 \\ 0.26 & 0.24 \end{bmatrix} z + \begin{bmatrix} 0 \\ 0.44 \end{bmatrix}$$

with probability 0.12.

You should accomplish this with a single if clause, although, you will want to make use of some subordinate elseif statements.

As an output for this part, produce a plot, where you use 4000 iterations and a fairly small markersize. Plot all points obtained through 4000 iterations. Include a title to your plot.

3 Project requirments

Write and submit both fern.m and fern.pdf files. The project will be graded using the follwoing rubric:

- 10 pts for header/program description,
- 10 pts for comments in code
- 10 pts for indentation in "for loop"
- 10 pts for indentation in "if else statments"
- 10 pts for correct code for part 1
- 10 pts for correct code for part 2
- 10 pts for correct plot for part 1 (2 pts for the title)
- 10 pts for correct plot for part 2 (2 pts for the title)
- 10 pts for the correct answer to the question in part 1
- 10 pts for submitting both .pdf and .m